

Assignment 5

COS221

Syntax Terror

University of Pretoria | Computer Science

Pretoria, May 2025



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Abstract



This assignment aims to provide a platform for which users can compare prices of various products across different retailers and suppliers, using the LAMP stack and a phpMyAdmin database.

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1

Introduction and Research

Authors: Syntax Terror

Official Repository:

<https://github.com/Sashumi6/COS221-Assignment5>

1.1 Understanding E-Commerce and Online Retail ^[6]

E-commerce has become an increasingly popular business model over the recent few years. E-commerce refers to the trade, or buying and selling, of goods and services over the Internet. Within e-commerce there are notable terms used interchangeably with it such as online retail and online shopping[7][8]. E-commerce can be categorised further into Business-to-Consumer, Business-to-Business, Business-to-Consumer, Consumer-to-Consumer, Consumer-to-Business and Consumer-to-Government. E-commerce is distinguishable by the use of personal devices and instant customer satisfaction.

1.2 Products bought by consumers, in retail and e-commerce

The products that are bought by consumers can be split into a few categories, such as: consumer goods, industrial goods and service goods[10]. Consumer goods being, as the name suggests, what you would buy yourself. While industrial is more reserved for companies and large processes[5][10]. Industrial goods can also be bought by general consumers but it is not as often and service being goods that are directly performed by other people. Even when looking at a tangible and intangible landscape, these three broad product categories still maintain their status. For intangibles, there are intangibles such as games and user software (logic pro x, photoshop etc.) but for industry there are also those such as company specific software to manage security and servers and service goods being consultants that work specifically with companies in order to get company specific tasks done. All these categories fit into both retail and ecommerce, and can be seen widely in their own sub categories in these places.

1.3 Price Comparison Tools^[9]

Price comparison websites or applications are platforms that assist customers in finding ideal prices for products they are interested in. These websites/applications allow the customer to compare prices across multiple retailers. These tools do not only assist consumers, but merchants as well by allowing them to attract new customers via good deals. Examples of these tools include BuyVia and Yahoo! Shopping, with both of these acting as price trackers that alert customers when there are price drops on products of interest. A tool that can assist merchants as well as customers is Google Shopping. Google Shopping helps merchants by providing them a report on competitor prices on products they have in common.

1.4 Content rating and categorization

Products and content on eCommerce sites are usually categorised and rated through a conjunction of automated means and user input. Artificial Intelligence and Machine Learning are often used to group products into different hierarchies (called product taxonomies) according to things such as type, brand and price^[2]. Customer reviews and ratings also play a large role in these autonomous technologies and can be used to give a more personal rating score which customers can trust and which can be made statistically accurate by using techniques such as weighted averages^[3]. Finally the guidelines and regulations of different companies can dictate how content is categorised by ensuring products and content are displayed and grouped in a legal, consistent and user-friendly manner^[4].

1.5 The Importance of user experience^[1]

User experience (UX) is important for reasons such as:

Usability; This refers to how users navigate and interact with the UI and the ease of access.

Credibility; A poorly designed UI can create a negative impression and drive users away. A UI with a sleek modern design and secure payment methods can instill confidence in users when using your website. Efficient Search & Navigation; having a robust system to search and navigate your website enhances the usability when elements, such as filters, are used so that users can have a productive and efficient way of finding what they need.

In other words, UX is important in the design of any eCommerce website as it attracts and imbues confidence in any user accessing the website which in turn will increase the chances a user will return to your site.

1.6 Popular Product Categories

In order to construct a realistic e-Commerce website that will appeal to the average online-shopper, this can be done by selling products that are most popular/sought-after by those who shop online. The Top 5 Online Shopping Categories Worldwide (By Revenue) in 2024 were "Electronics", "Fashion", "Food", "Beverages" and "DIY/hardware" items.[11] Another source provides the following "Top-5" list for 2024 based on popularity: "Electronics and Technology", "Fashion and Apparel", "Health and Beauty", "Home and Living" and "Groceries and Food Delivery".[12] The Top 5 categories on a list for 2022-2023 were "Electronics and Gadgets", "Fashion and Apparel", "Home/Kitchen Appliances", "Health and Beauty" and "Toys and Games".[13] Over the last three years a common theme can be observed that Electronics, Fashion, Beauty/Health and Home/Hardware appliances are the most popular options amongst online-shoppers and should therefore be considered as viable options to be stocked on our e-Commerce website in terms of realistically being able to appeal to customers.

2

(E) E-R Diagram

Below is our (E) E-R Diagram.

Figure 1. (E) E-R Diagram

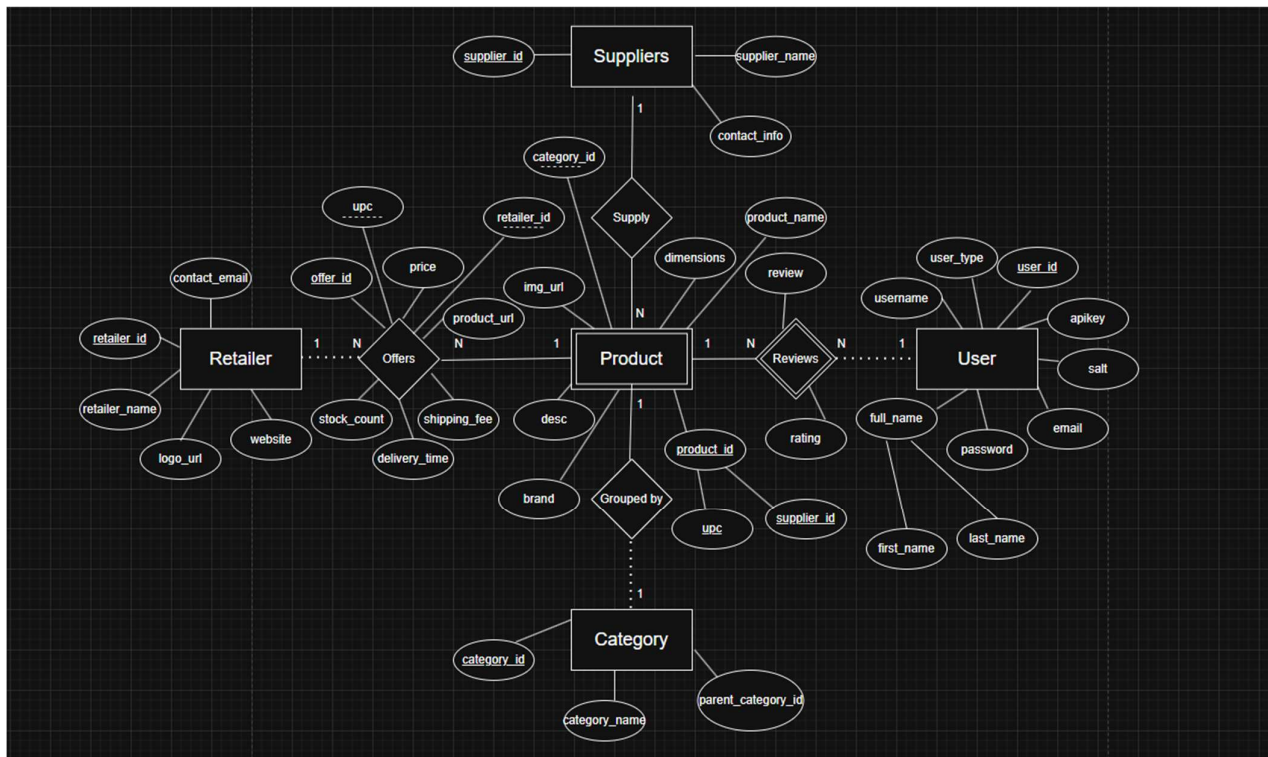
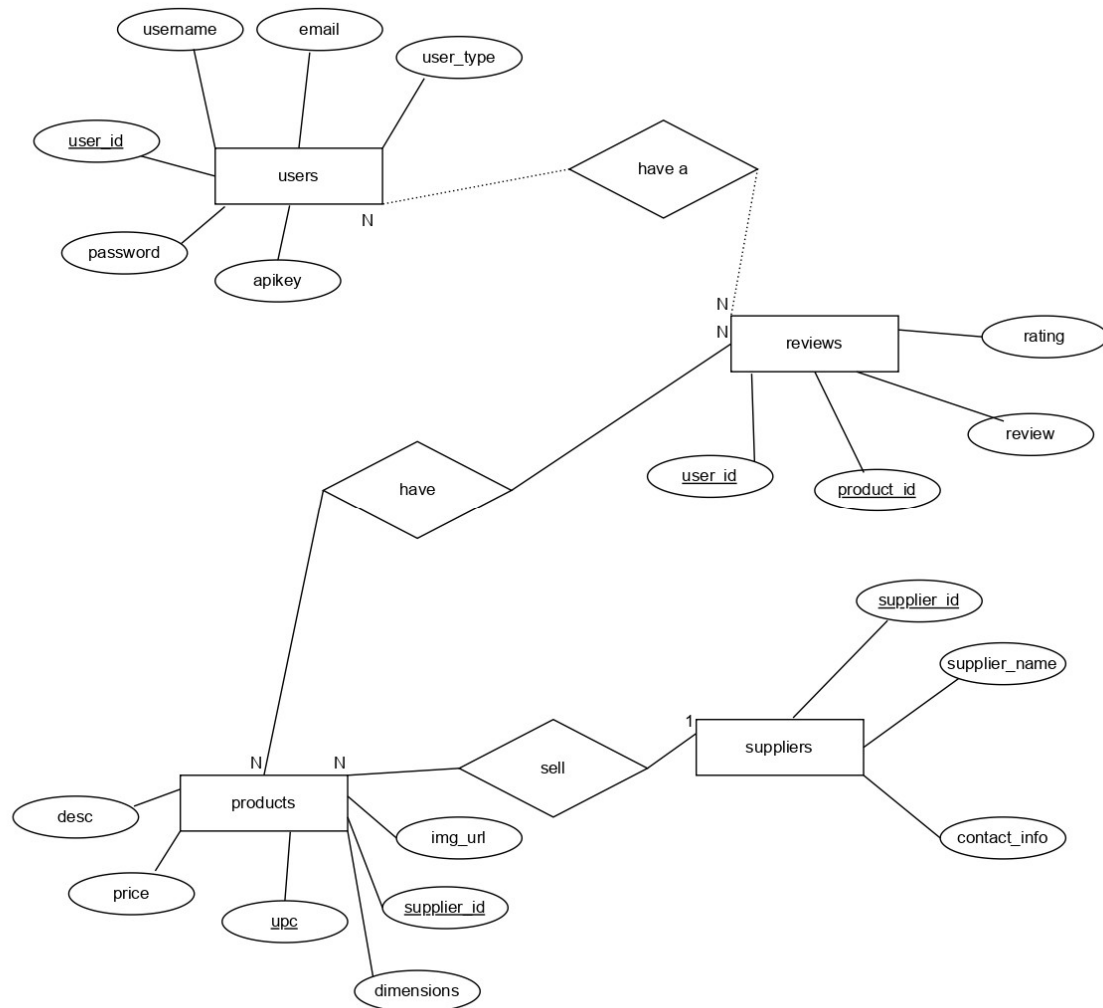


Figure 1. Final (E) E-R Diagram

Note

Figure 1. refers to the final iteration of the E-ER Diagram please find the initial iteration.

Figure 2, below



2.2 Iterations and improvements

The initial iteration of the (E) E-R diagram provides a much simpler design with fewer relationships and entities, however this presents problems as it makes the system quite limited and does not allow for robust data management and manipulation. The final iteration of the database table provides far more flexibility through the addition of new entities such as Retailer and Category and relationships such as offers, as well as through recursive identification. ie. category_id and parent_category_id in the Category entity.

2.3 Assumptions

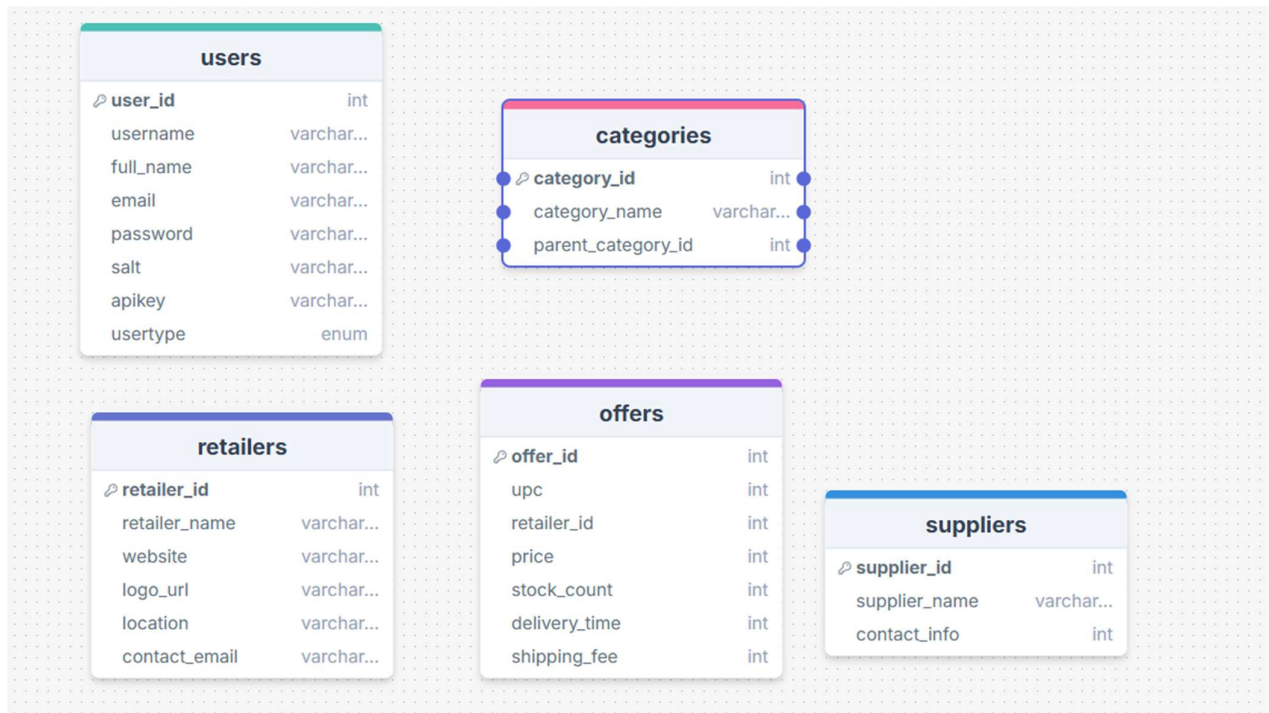
We firstly have to assume that each product has a upc (unique product code) which uniquely identifies each product.

3

Mapping

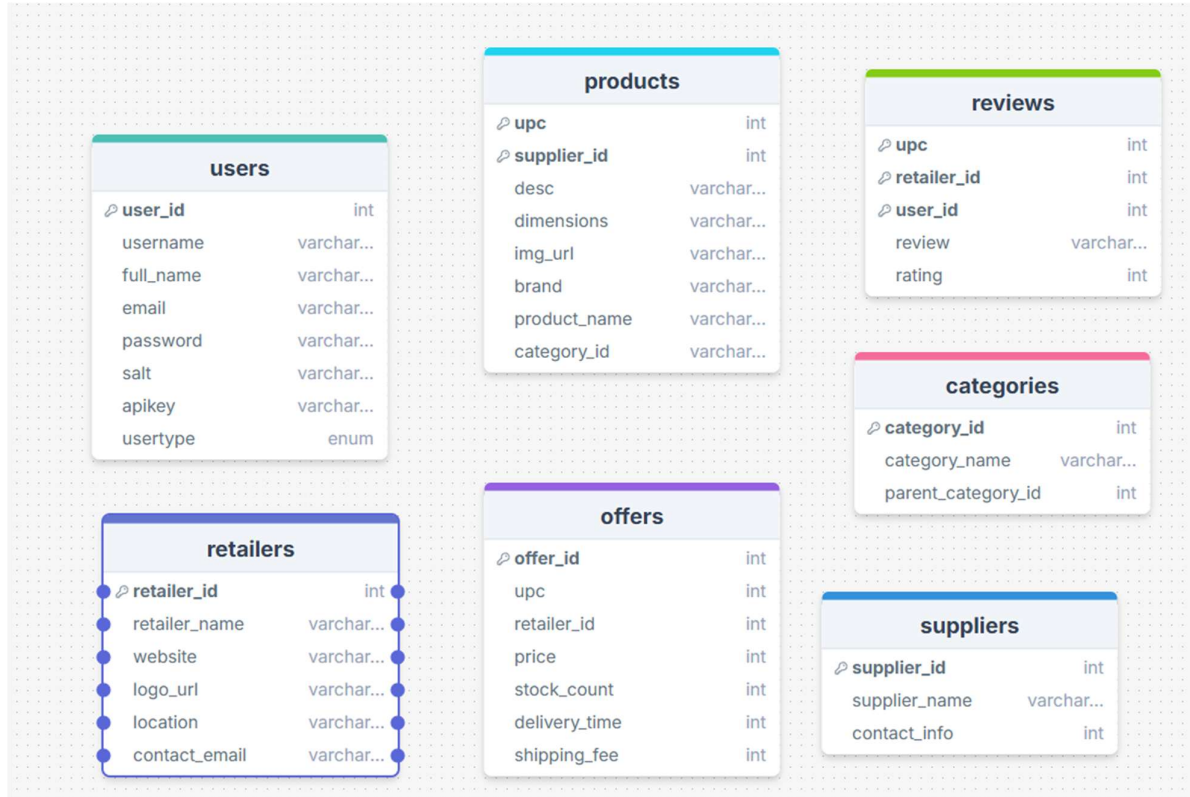
3.1 Regular entities

Our database has 5 regular entities: users, categories, retailers, offers and suppliers.



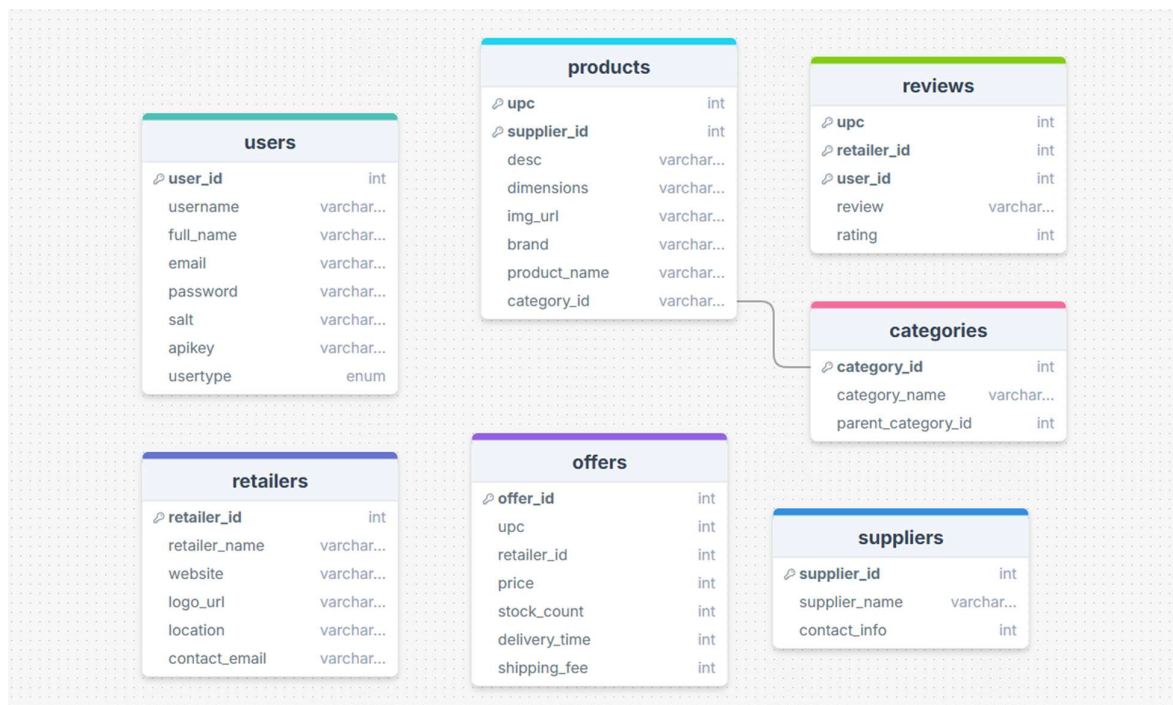
3.2 Weak entities

The database has two weak entities: reviews and products.



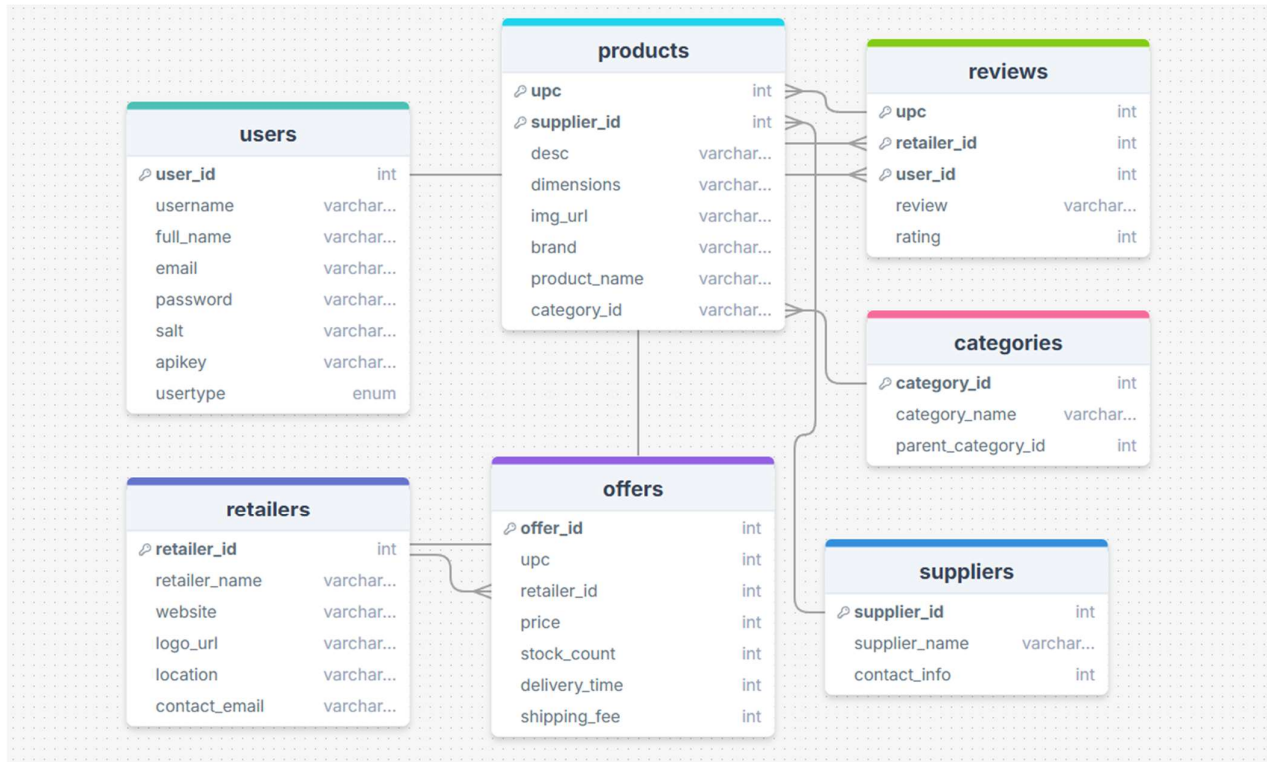
3.3 1:1 relationship

Our database has one 1:1 relationship.



3.4 1:N relationships

Our database has multiple 1:N relationships.



3.5 M:N relationships

Our database has no M:N relationships.

3.6 Multivalued attributes

Our database has no Multivalued attributes.

3.7 N-ary relationships

Our database has no N-ary relationship.

3.8 Specialization, generalization and unions

Our (E) E-R diagram had no specialization, generalization or unions.

4

Analyze & Optimize

The Query that we have chosen to analyze is as shown below, which takes 0.0024s (upc = 1001 as placeholder data)

```
SELECT u.username, r.review, r.rating, s.supplier_name
FROM reviews r JOIN users u ON u.user_id = r.user_id
JOIN suppliers s ON s.supplier_id = r.supplier_id
WHERE r.upc = 1001;
```

Query 1

This query has one major flaw which is that only *after* all the data has been joined does it then search for that specific upc, and so to optimize it one should firstly fetch all tuples with upc = 1001 and then join the respective tables.

```
SELECT u.username, r.review, r.rating, s.supplier_name
FROM
  (SELECT * FROM reviews WHERE upc = 1001) r
INNER JOIN users u ON u.user_id = r.user_id
INNER JOIN suppliers s ON s.supplier_id = r.supplier_id;
```

Query 2

This optimized query takes 0.0023s to complete which is because there isn't sufficient data to experimentally deduce a gain/loss percentage between the two queries. But theoretically Query 2 is 46% faster than Query 1 with the average case, although this is dependent.

The reason why Query 2 is faster than Query 1 is because instead of joining the tables first, thus joining redundant tuples (tuples which do not have the upc we want), we first filter out unwanted tuples and then join tuples after which can effectively half computation time. For example, if we have 10,000 tuples and only 200 of which have a upc = 1001 we do not need to waste time joining values which will not be included in our desired resultant tuples.

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